MODULE 6 · OPERATIONS AND HUMAN FACTORS

Lesson 29: Regulations, Standards and Safety Management

Lesson 29 of 30 · about 8 min

LEARNING OBJECTIVES

- Identify the main international regulations and standards for cold weather operations.
- Describe the elements of a ground de-icing/anti-icing programme.
- Use the safety management system to report winter hazards.



ICAO Doc 9640 and Circular 355 — part of the international framework.

International framework

Source	Content
ICAO Annex 6, Part I	Operator responsibilities, including the requirement that an aircraft is not to take off unless it has been inspected for icing and, if necessary, de-iced and anti-iced.
ICAO Doc 9640	Manual of Aircraft Ground De-icing/Anti-icing Operations.
ICAO Annex 14 and PANS-Aerodromes (Doc 9981)	Global Reporting Format for runway surface conditions.
ICAO Doc 10064	Aeroplane Performance Manual, including contaminated runway performance.
EASA Air Operations	CAT.OP.MPA.250 (ice and other contaminants — ground procedures) and CAT.OP.MPA.255 (in-flight procedures).
FAA	14 CFR 121.629 (Clean Aircraft Concept and de-icing programme) and AC 120-60 (ground de-icing/anti-icing programme).

Transport Canada / FAA	Holdover time guidelines, published every winter season.
SAE International	AS6285 (de-icing/anti-icing processes), AS6286 (training and qualification), AMS1424 and AMS1428 (fluid specifications).

National regulations from the civil aviation authority, the aircraft AFM, and the Domestic Airlines Operations Manual implement these standards for our operation. They are the documents you must follow.

The ground de-icing/anti-icing programme

Operators must have an approved programme that typically includes:

- responsibilities of the operator, the commander and the service providers;
- the procedures for inspection, treatment, communication and the anti-icing code;
- the holdover time guidelines to be used, and how to use them;
- initial and recurrent training for flight crew, cabin crew, dispatchers and ground staff;
- quality control of fluids, equipment and service providers;
- the pre-take-off check and pre-take-off contamination check procedures.

Safety management and reporting

Winter operations produce many precursors to accidents. Report them through the company safety reporting system:

- contamination found after de-icing, or a take-off with suspected contamination;
- communication errors with de-icing crews or ATC;
- fluid smells or smoke in the cabin;
- braking action worse than reported, and runway excursions or near-excursions;
- control stiffness or restriction, and residue findings;
- holdover times exceeded, or pressure to depart.

Domestic Airlines operates a **just culture**: honest reports of errors and hazards are used to learn and to improve the system, not to blame individuals.

KEY TAKEAWAYS

- ICAO, EASA, FAA, Transport Canada and SAE set the international framework; national rules, the AFM and the Operations Manual apply it.
- Operators need an approved ground de-icing/anti-icing programme with training and quality control.
- Report winter hazards and errors — just culture protects honest reporting.

KNOWLEDGE CHECK

Which organisation publishes the holdover time guidelines used worldwide each winter?

A. IATA and ICAO

B. The FAA and Transport Canada

C. Each airport authority

D. The aircraft manufacturer only

Answer: B. Holdover time guidelines are published each season by the FAA and Transport Canada and are adopted by operators worldwide.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.